

JAEYOUNG KIM

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EDUCATION

8/2022 - present	Graduate Research Assistant Homogeneous Dynamics (PI: Seonhee Lim)	Seoul National University (SNU)
3/2022 - present	Integrated MA/Ph.D Course Master Course Mathematical Sciences	SNU
6/2020 - 3/2021	Research Internship Quantum Information Theory (PI: Sang-Gyun Youn)	SNU
3/2016 - 2/2022	Bachelor of Science Mathematical Education Leave for Mandatory Military Service (1/2018 - 9/2019)	SNU

PUBLICATION AND PREPRINTS

- Euijung Chang, Jaeyoung Kim, Hyesun Kwak, Hun Hee Lee, Sang-Gyun Youn (2022). Irreducibly $SU(2)$ -covariant quantum channels of low rank. *Reviews in Mathematical Physics*, 34(07), 2250021.
- Jaeyoung Kim, Sanghoon Kwon, Seonhee Lim (2026+). Markov chain of diagonal action on the standard quotient of the building associated to $PGL_3(\mathbb{F}_q((t^{-1})))$. *Work in progress*.
- Jaeyoung Kim(2026+). Distribution of periodic torus orbits on Siegel modular variety and Duke's theorem for matrix-valued quadratic form. *Work in progress*.

TALKS

06/2025	Markov chain of diagonal action on the standard quotient of the building associated to $PGL_3(\mathbb{F}_q((t^{-1})))$ Poster Session, Beyond Uniform Hyperbolicity, Abdus Salam International Centre for Theoretical Physics (ICTP) Trieste, Italy
10/2024	Markov chain of diagonal action on the standard quotient of the building associated to $PGL_3(\mathbb{F}_q((t^{-1})))$ Special Session, 2024 KMS Annual Meeting, Korean Mathematical Society (KMS) Suwon, South Korea
08/2024	Dynamical property for horospherical action on the standard quotient of the building $\mathcal{B}_3(\mathbb{F}_q((t^{-1})))$ Lightning Talk, Dynamical Group Theory III: New Methods in Group Actions on Manifolds, Korea Institute for Advanced Study (KIAS) Seoul, South Korea
04/2024	Dynamical property for horospherical action on the standard quotient of the building $\mathcal{B}_3(\mathbb{F}_q((t^{-1})))$ Lightning Talk, Introductory school: Rigidity, Dynamics and Geometric Structures, Centre International de Rencontres Mathématiques (CIRM) Marseille, France
10/2023	Non-escape of mass for horospherical action on the standard quotient of the building $\mathcal{B}_3(\mathbb{F}_q((t^{-1})))$ under type 1 geodesic flow Poster Session, 2023 KMS Annual Fall Meeting, KMS SNU, Seoul, South Korea

PRESENTATIONS AND SEMINAR TALKS

- Work of Lindenstrauss, Margulis, Mohammadi, Shah: Effective Avoidance Principle and Diophantine Condition*, Research Seminar on Homogeneous Dynamics, April 2025.
- Work of Cantat, DuJardin: Hyperbolic dynamics on Compact Kähler Surfaces*, Course: Topics in Topology, December 2024.
- Quasi-Fuchsian groups and Bers' Simultaneous Uniformization theorem*, SNU Graduate Course: Differential Geometry 2, November 2024.
- Sullivan's No Wandering Domain Theorem*, SNU Graduate Course: Differential Geometry 2, October 2024.
- Markov chain of diagonal action on the standard quotient of the building associated to $PGL_3(\mathbb{F}_q((t^{-1})))$* , Exchange Seminar, October 2024.
- Iteration of Polynomial-like maps*, SNU Graduate Course: Differential Geometry 2, September 2024.

- *The Scope of Dirichlet forms: Closability and Examples*, Course: Topics in Mathematical Methods of Probability, April 2024.
- *Bruhat-Tits tree of SL_2 over a local field*, Research Seminar on Homogeneous Dynamics, SNU, March 2024.
- *Work of Ciobotaru, Finkelshtein, Sert: Non-escape of mass and Equidistribution for Horospherical actions on Trees*, Exchange Seminar, SNU, August 2023.
- *Work of Michel, Venkatesh: Equidistribution, L-functions and Ergodic theory*, Course: Topics in Topology, June 2023.

SEMINARS

01/2023 - present	Exchange Seminar Supervisor: Gyeseon Lee	SNU
09/2023 - 08/2024	Reading Seminar on Arithmetic of Hyperbolic 3-manifolds	SNU
09/2022 - present	Research Seminar on Homogeneous Dynamics Supervisor: Seonhee Lim	SNU

CONFERENCES AND WORKSHOPS ATTENDED

- **2026 Algebra Camp**, Chuncheon, South Korea, January 2026.
- **Dynamical Group Theory VII**, SNU, Seoul, South Korea, November 2025.
- **2025 KMS Annual Meeting**, KMS, Seoul, South Korea, October 2025.
- **Dynamical Group Theory VI**, KIAS, Seoul, South Korea, September 2025.
- **Numeration and Substitution 2025**, University of Tsukuba, Japan, September 2025.
- **2025 2K-GATE One day Geometric Topology Festival**, KIAS, Seoul, South Korea, August 2025.
- **A Summer Dynamics Day**, Dongguk University, Seoul, South Korea, August 2025.
- **One-day Festival of Group actions, Geometry and Dynamics**, KIAS, Seoul, South Korea, August 2025.
- **Geometry and Dynamics in Higher Rank Lie Groups**, SLMath, Berkeley, USA, July 2025.
- **Beyond Uniform Hyperbolicity**, ICTP, Trieste, Italy, June 2025.
- **2025 KMS Spring Meeting**, KMS, Daejeon, South Korea, April 2025.
- **Workshop on Homogeneous and Complex Dynamics 2025**, SNU, Seoul, South Korea, March 2025.
- **KIAS-LFANT Winter School on Number Theory**, KIAS, Seoul, South Korea, January 2025.
- **2024 KMS Annual Meeting**, KMS, Suwon, South Korea, October 2024.
- **Dynamical Group Theory IV: KIAS-Rice Workshop on Geometric Topology**, June E Huh Center for Mathematical Challenges (HCMC) of KIAS, Seoul, South Korea, September 2024.
- **Dynamical Group Theory III: New Methods in Group Actions on Manifolds**, KIAS, Seoul, South Korea, August 2024.
- **Number Theory Intensive Lecture Series: Introduction to p-adic Hodge theory**, POSTECH, Pohang, South Korea, June 2024.
- **Introductory school: Rigidity, Dynamics and Geometric Structures**, CIRM, Marseille, France, April 2024.
- **2024 Algebra Camp**, Yangpyeong, South Korea, February 2024.
- **Winter School on Homogeneous Dynamics**, Shanghai Center for Mathematical Sciences, Fudan University, Shanghai, January 2024.
- **Hyperbolic Geometry of Numbers**, HCMC of KIAS, Seoul, December 2023.
- **2023 KMS Annual Fall Meeting: KMS Special Conference with 2022 Fields Medalists**, KMS, Seoul, October 2023.
- **Summer School on Arithmetic Statistics**, KIAS, Seoul, August 2023.
- **2023 KMS Spring Meeting**, KMS, Daejeon Convention Center, Daejeon, Korea, April 2023.
- **Workshop on Dynamics of Group Action**, Gangneung, South Korea, April 2023.
- **QSMS 2023 Winter Workshop on Number Theory and Representation Theory**, QSMS, Lahan Hotel Jeonju, Jeonju, Korea, January - February 2023.

TEACHING

Spring & Fall 2025	TA: Differential and Integral Calculus Practice 1 (003) & 2 (005)	SNU
Spring & Fall 2024	TA: Calculus Practice 2 (001, 026)	SNU
Spring 2023	TA: Honor Calculus Practice 2 (006)	SNU
Spring 2023	Grader: Number Theory (001)	SNU
Spring 2023	TA: Calculus Practice 1 (021)	SNU
Spring & Fall 2022	TA: Calculus Practice 1 (022) & 2 (009)	SNU
05/2021	Teaching Practice	Seoul National University Middle School
Summer & Winter 2020	Teaching Volunteer	Guhyun High School

HONORS

03/2022 - present	Graduate Student Instructor Scholarship	SNU
03/2022 - 02/2023 09/2023 - 02/2025 09/2025 - present	BK21 Reasearch Fellowship	SNU
03/2023 - 08/2023	Next-Generation Scholars in Fundamental Academic Disciplines	SNU

LANGUAGES

English - B2, **Korean** - native